

Restricted Boltzmann Machines on MNIST: contrastive divergence, energy barriers and hyperparameter sensitivity

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Abstract

A Restricted Boltzmann Machine (RBM) is a two-layer Boltzmann distribution that learns by playing a tug-of-war between the data it sees and the fantasies it samples. We train one with $L=12$ Bernoulli hidden units on the binarised MNIST sub-dataset of digits $\{0, 1, 2\}$ and use it as a controlled laboratory for the four ingredients that make energy-based learning work: contrastive divergence (CD), Hinton’s visible-bias initialisation, the quasi-ergodic structure of the Gibbs sampler, and hyperparameter sensitivity. Reading the gradient root-mean-square (RMS) as a thermometer for “how much real signal is left”, we measure the irreducible $\sim 1/\sqrt{N}$ Monte-Carlo noise floor and confirm that training becomes sampling-noise-limited after ~ 50 epochs. Forced pixel-flip transitions reveal free-energy basins separated by barriers $\Delta F \sim 60\text{--}100$ (SNR = 1.1–1.8), suppressing spontaneous CD class-mixing by $10^{-28}\text{--}10^{-46}$. A ten-configuration sweep ranks the hidden width L , the L1 weight γ , and the CD depth N_t as the dominant knobs, with RMSprop providing robust optimisation with little tuning. Each observation is tied to a concrete sampling-theoretic or information-theoretic mechanism, and we propose a single-number quality metric, $\Delta F/\sigma_{\text{within}}$, as a sharper diagnostic than the visual quality of generated digits.

1. INTRODUCTION

A Restricted Boltzmann Machine (RBM) [1, 2, 3] is a two-layer, undirected, energy-based generative model. It places a Boltzmann distribution over a vector of visible units $\mathbf{v}$ – which carry the data – and a vector of hidden units $\mathbf{h}$, which the model uses to invent a compact latent description of that data. The “restriction” is simply the absence of intra-layer edges: hidden units do not talk to other hidden units, and visible units do not talk to other visible units. This structural choice has a powerful pay-off, because the conditional distributions $P(\mathbf{h}|\mathbf{v})$ and $P(\mathbf{v}|\mathbf{h})$ become products over independent units, so sampling a whole layer is one vectorised matrix multiply followed by a Bernoulli draw – block Gibbs sampling. Although the partition function still involves an intractable sum over 2^{D+L} states, the gradient of the log-likelihood reduces to the difference of two simple correlation functions whose “data” phase is exact and whose “model” phase is approximated by Contrastive Divergence (CD).

The aim of this work is twofold. First, we want to verify, on a non-trivial real dataset, the chain of theoretical statements that motivates the algorithm: does the gradient of the log-likelihood actually decay to a predictable Monte-Carlo noise floor; does Hinton’s logit initialisation of the visible bias really place the model marginal on top of the data marginal before training begins; does the trained model carve clearly identifiable class basins in free-energy space. Second, we ask the comparative question: among the many knobs of RBM training – optimiser, encoding, regularisation strength, hidden capacity, CD depth – which ones really matter once the others are sensibly chosen.

We restrict attention to MNIST digits $\{0, 1, 2\}$, binarised at 50% intensity, because this sub-dataset is rich enough to display the full phenomenology (multiple basins, stroke detectors, class templates, dead units) yet small enough to permit ten controlled training runs. The investigation is steered by four diagnostics introduced in Sec. 2 and applied systematically in Sec. 3.

2. METHODS

Energy and probability. With visible biases $\mathbf{a} \in \mathbb{R}^D$, hidden biases $\mathbf{b} \in \mathbb{R}^L$, and weights $W \in \mathbb{R}^{D \times L}$, the joint energy and Boltzmann distribution read

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^\top \mathbf{a} - \mathbf{h}^\top \mathbf{b} - \mathbf{v}^\top W \mathbf{h}, \quad (1)$$

$$P(\mathbf{v}, \mathbf{h}) = Z^{-1} e^{-E(\mathbf{v}, \mathbf{h})}, \quad Z = \sum_{\mathbf{v}, \mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}. \quad (2)$$

Bipartiteness gives the factorised conditionals

$$P(h_j = 1 | \mathbf{v}) = \sigma\left(b_j + \sum_i W_{ij} v_i\right),$$

$$P(v_i = 1 | \mathbf{h}) = \sigma\left(a_i + \sum_j W_{ij} h_j\right), \quad (3)$$

with $\sigma(u) = (1 + e^{-u})^{-1}$ the logistic function. Marginalising over the hidden layer yields a free energy that makes $P(\mathbf{v}) \propto e^{-F(\mathbf{v})}$ exact and cheap to compute – no hidden-side sampling needed:

$$F(\mathbf{v}) = -\mathbf{v}^\top \mathbf{a} - \sum_{j=1}^L \log(1 + e^{H_j(\mathbf{v})}), \quad (4)$$

with the local field $H_j(\mathbf{v}) = b_j + \sum_i W_{ij} v_i$. Equation (4) is the Lego brick of every diagnostic in Sec. 3: it lets us score any visible image without ever sampling the hidden layer.

Gradient and CD- k . Maximising $\log P(\mathbf{v})$ over the data gives the canonical positive–negative phase decomposition

$$\partial_{W_{ij}} \log P(\mathbf{v}) = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}. \quad (5)$$

Intuitively, the positive phase pushes the energy of real data *down*, the negative phase pushes the energy of the model’s fantasies *up*; learning halts when the two are indistinguishable. The first term is exact (one stochastic hidden sample per data point); the second is approximated by initialising a Gibbs chain on a data point and running it for k alternating steps, $\mathbf{v}_{\text{data}} \rightarrow \mathbf{h}_0 \rightarrow \mathbf{v}_1 \rightarrow \dots \rightarrow (\mathbf{v}_k, \mathbf{h}_k)$. We use $k = N_t = 2$ as the baseline, with $N_t \in \{1, 5\}$ tested in the sweep.

Hinton’s bias initialisation. Setting $W = 0$ and $\sigma(a_i) = p_i$, with p_i the empirical on-pixel frequency, gives the unique value of $\mathbf{a}$ that makes the model marginal coincide with the data marginal at epoch zero:

$$a_i = \sigma^{-1}(p_i) = \log \frac{p_i}{1 - p_i}, \quad (6)$$

clipped to $p_i \in [\epsilon, 1 - \epsilon]$, $\epsilon = 10^{-4}$, to keep the logit finite on always-on or always-off pixels. The intuition is the cleanest possible bias prior: before any weight has been updated, every pixel is already turned on by the model with the correct frequency, so the entire learning budget can be spent on the much more informative weight gradient [3].

Optimiser, regularisation and minibatches. Updates use RMSprop ($\beta = 0.9$, $\varepsilon = 10^{-4}$, $\eta = 0.05$): a per-parameter adaptive step that automatically damps directions of large gradient variance – a free benefit on the heteroscedastic CD gradient. An ℓ_1 penalty ($\gamma = 10^{-3}$) shrinks small weights exactly to zero, producing compact filters. Minibatches grow quadratically from $N_{\text{ini}} = 10$ to $N_{\text{fin}} = 500$ over 150 epochs (20 batches/epoch): noisy batches early for exploration, large batches late whose Monte-Carlo noise scales as $1/\sqrt{N}$ – the *noise floor* we use as the reference in Sec. 3.

Diagnostics. We monitor (i) the per-epoch gradient RMS, $\text{std}(g_W), \text{std}(g_a), \text{std}(g_b)$, against the $1/\sqrt{N}$ floor; (ii) the receptive fields $W_{:,j} \in \mathbb{R}^{784}$ reshaped as 28×28 images; (iii) free Gibbs chains of length 80–300 to test generative quality; (iv) free-energy profiles $F(\mathbf{v}_t)$ along forced pixel-flip transitions between class endpoints, quantified by the barrier $\Delta F = \max_t F(\mathbf{v}_t) - \frac{1}{2}[F(\mathbf{v}_{\text{start}}) + F(\mathbf{v}_{\text{end}})]$ and the SNR $\Delta F/\sigma_{\text{within}}$, where σ_{within} is the pooled within-class standard deviation of F . Statistical uncertainty comes from $N_{\text{pairs}} \times N_{\text{rep}}$ independent endpoint pairs and random orderings, summarised by 95% confidence intervals.

3. RESULTS

3.1. Filters and generative samples

Figure 1 summarises the trained model. The visible bias $\mathbf{a}$ – initialised by Eq. (6) – is dark (strongly negative) at the image border, where pixels are almost never on, and lighter near the centre, where they often are; before any weight update the model marginal $\sigma(a_i)$ matches the empirical frequency p_i to within the clipping tolerance 10^{-4} . This is exactly the bias-prior benefit of Hinton initialisation: the gradient on $\mathbf{a}$ starts near zero, so the optimiser is free to spend its first epochs sculpting W , where the actual structure of digits lives.

The twelve learned filters separate cleanly into two populations. A handful of units (e.g. h_1, h_2, h_5) acquire whole-digit templates: their weight images strongly resemble silhouettes of the digit “1” (vertical bar of excitatory weights flanked by inhibitory regions). Other units (e.g. h_4, h_6, h_8, h_9) become stroke detectors: their receptive fields display localised arcs that respond to either the upper or lower curvature of a “0” or the diagonal of a “2”. Filter h_3 has been collapsed to $W_{:,3} \approx 0$ by the ℓ_1 penalty – a dead unit, the visible signature of redundant capacity (cf. Sec. 3.4). The generative panel in Fig. 1(d) confirms

that the model is not just a static classifier of features: free Gibbs chains converge to recognisable digits within a few tens of steps and remain realistic afterwards.

3.2. Convergence and the noise floor

Figure 2 shows the per-epoch RMS of the three gradients on a log–log scale. All three decay, but at very different rates: the visible bias gradient decays the fastest (Hinton initialisation already places it near optimum and noise dominates almost immediately); the hidden bias gradient is the most volatile (it is just a single number per unit and therefore a factor $\sqrt{L/D} \sim 8$ noisier than the weight gradient); and the weight gradient decays as a clean power law and approaches the $1/\sqrt{N}$ reference at the end of training. The match with the noise floor is the operationally meaningful definition of convergence here: continuing to train without enlarging N (or annealing η) cannot reduce $\text{std}(g_W)$ further because the CD estimator itself is no more accurate than that. The quadratic minibatch schedule $N(\text{epoch})$ deliberately exploits this: it lowers the noise floor itself in the late phase, allowing the model to fine-tune past where a fixed- N schedule would stall.

3.3. Energy barriers and quasi-ergodicity

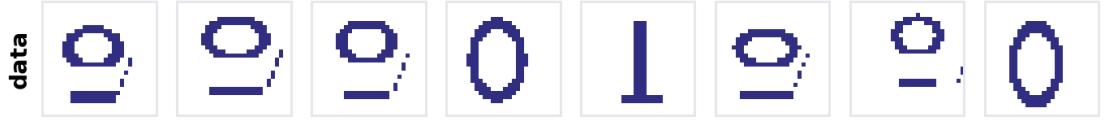
The most direct way to ask whether the model has learned class structure is to evaluate F along an explicit path between two examples of different classes. We pick $\mathbf{v}_{\text{start}}$ from class A and $\mathbf{v}_{\text{end}}$ from class B , build $\mathbf{v}_t$ by flipping a fraction t/T of the differing pixels in a randomised order, and use the exact free-energy formula (4) at each step. Figure 3 shows the result, averaged over many endpoint pairs and pixel orderings. All three transitions exhibit a pronounced ridge in the middle of the path: the model assigns much higher free energy to images that are half digit- A and half digit- B than to either endpoint – the same landscape geometry as a first-order phase transition between two ferromagnetic phases, with the digit basins playing the role of the two pure-phase minima. Quantitatively, the barrier heights ΔF are 104, 62 and 84 for the pairs $0 \rightarrow 1$, $0 \rightarrow 2$, $1 \rightarrow 2$ respectively, with a pooled $\sigma_{\text{within}} \approx 59$, giving SNR = 1.8, 1.1, 1.4: the barriers exceed within-class fluctuations and are statistically significant. Because $P(\mathbf{v}) \propto e^{-F(\mathbf{v})}$, an unbiased Gibbs chain visits the saddle with probability suppressed by $e^{-\Delta F} \sim 10^{-28} \dots 10^{-46}$. This is the concrete origin of the *quasi-ergodic* behaviour of CD: the chain does not violate ergodicity in principle, but in practice it is exponentially trapped, and a finite-time run almost never crosses the ridge.

The barrier picture predicts that free Gibbs chains, started at any data point, should remain trapped in their basin. Figure 4 verifies this directly: chains started on a “1” stay deep in the blue basin, those started on a “0” or “2” stay near the corresponding horizontal stripe, and only $\leq 0.6\%$ of all CD steps ever leave the starting band. The qualitative behaviour of the sampler is therefore not a pathology of any single chain but a consequence of the free-energy landscape itself.

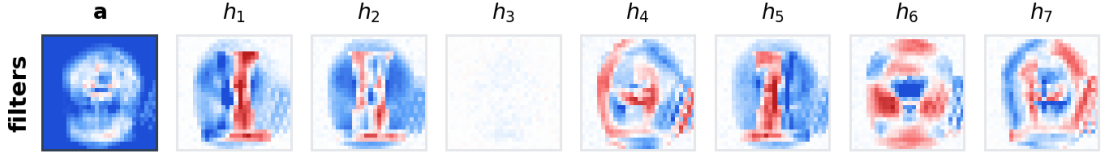
3.4. Hyperparameter sensitivity

Figure 5 summarises the ten-configuration sweep. Each panel measures one of the three quantitative diagnostics; we highlight the four conclusions that emerge

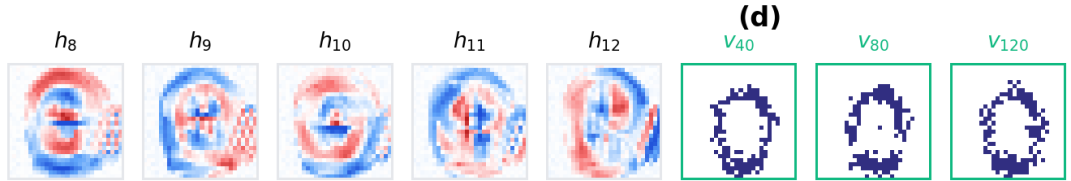
(a)



(b)



(c)



(d)



Figure 1. End-of-training panorama for the baseline RBM (MNIST $\{0, 1, 2\}$, $L = 12$, $N_t = 2$, RMSprop, $\gamma = 10^{-3}$). (a) Eight binarised data points fed to the model. (b)–(c) Visible bias $\mathbf{a}$ (leftmost panel of (b), highlighted in dark slate) and the twelve learned receptive fields $W_{:,1..12}$. Red weights are excitatory (ink there drives $h_j \rightarrow 1$), blue inhibitory. Two distinct populations of features emerge: *digit templates* (e.g. the central bar of h_1 , h_2 and h_5 which detects the stroke of a “1”) and *stroke detectors* (the curved-arc weights of h_4 , h_6 , h_8 , h_9 which match the upper and lower arcs of a “0” or “2”). Filter h_3 has been driven to zero by the ℓ_1 penalty – a sparse, dead unit, the visible signature of redundant capacity. (d) Three frames from a free Gibbs chain at $t = 40, 80, 120$: the model dreams recognisable digits, confirming that it has learned an entire distribution $P(\mathbf{v})$ rather than a point estimate.

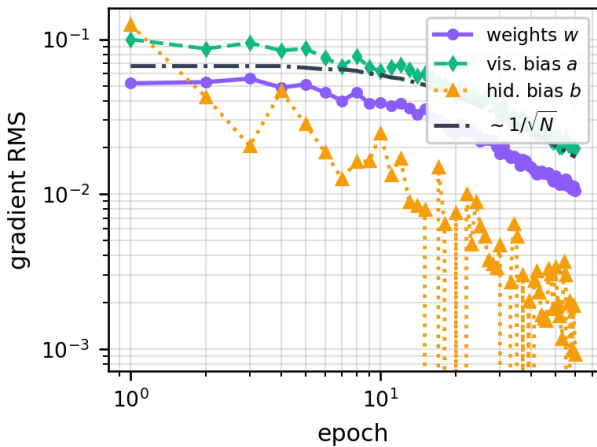


Figure 2. Gradient RMS $\text{std}(g)$ as a function of training epoch (log–log). All three parameter groups decay roughly as power laws. The dash-dotted reference $\sim 1/\sqrt{N}$ tracks the irreducible Monte-Carlo noise of a CD estimate at minibatch size N . Once $\text{std}(g_W)$ touches that curve, further epochs mostly chase noise – enlarging N or shrinking η becomes more profitable than running more epochs.

consistently across them.

(i) *Optimiser.* RMSprop reaches a final gradient RMS within a factor of ~ 2 of plain SGD with a much higher learning rate, but it does so with negligible tuning effort: the per-parameter adaptive step automatically absorbs the heteroscedastic CD noise. This makes RMSprop the

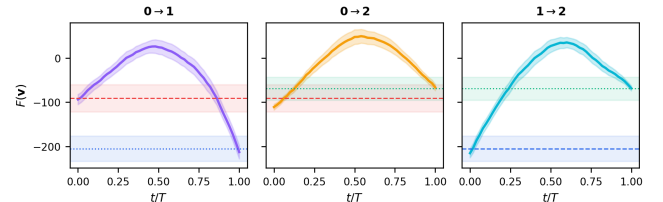


Figure 3. Free-energy profiles $F(\mathbf{v}_t)$ along forced pixel-flip transitions between digit pairs ($N_{\text{pairs}} \times N_{\text{rep}} = 32$ independent paths per pair, shaded band = 95 % CI). Horizontal stripes mark the within-class $\langle F \rangle \pm \sigma$ for the start (dashed) and end (dotted) digits. Each transition crosses a clear ridge of ~ 60 – 100 free-energy units, i.e. ~ 1 – 2 within-class standard deviations. The Boltzmann factor $e^{-\Delta F}$ then suppresses spontaneous Gibbs class transitions by $\sim 10^{-28}$ to 10^{-46} .

safer default choice when the noise structure of the gradient is unknown a priori.

(ii) *Encoding.* The SPINS encoding $\{-1, +1\}$ doubles the *level_gap* and therefore multiplies every conditional logit by 2; without simultaneously halving η this drives the optimiser into a poorly-conditioned regime, visible in panel (a) as a final RMS an order of magnitude above all other runs. POTS (one-hot hidden) is the opposite extreme: mutual inhibition forces each unit to specialise, the gradient RMS becomes the smallest in the sweep, but capacity is reduced to a single L -state categorical hidden variable.

(iii) *Regularisation.* The ℓ_1 weight γ controls the trade-off between filter sharpness and feature richness.

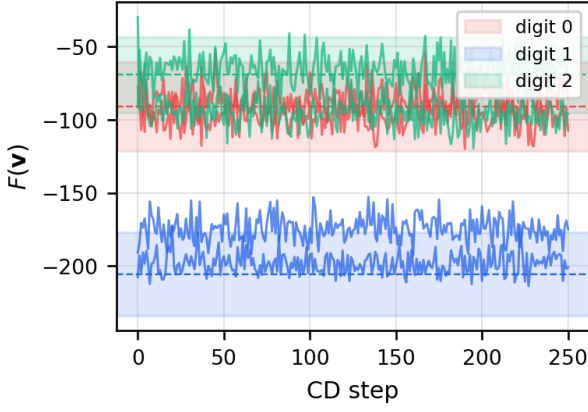


Figure 4. Free Gibbs/CD chains of 250 steps started inside each class basin (two chains per class). The chains never leave the band of their starting class, even though the basins are only $\sim 1\text{--}2 \sigma_{\text{within}}$ apart in F . The crossing fraction is $\leq 0.6\%$ for all three starting classes – the experimental fingerprint of the energy barriers in Fig. 3.

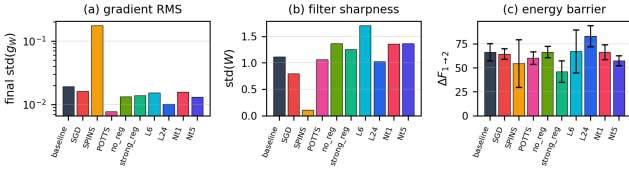


Figure 5. Ten configurations, each changing one knob away from the baseline (RMSprop, bits, $L = 12$, $N_t = 2$, $\gamma = 10^{-3}$), trained for 50 epochs. (a) Final gradient RMS $\text{std}(g_W)$ (log axis): RMSprop variants cluster $\sim 10^{-2}$ while SPINS is an order of magnitude noisier (the $\times 2$ energy gap blows up the effective learning rate). (b) Filter sharpness $\text{std}(W)$: over-shrinkage (strong_reg) and SPINS produce nearly flat filters, while $L=6$ packs more signal per unit. (c) Free-energy barrier $\Delta F_{1 \rightarrow 2}$ with 95% CI; capacity (L24) gives the highest barrier, strong ℓ_1 erodes it.

$\gamma = 0$ (no_reg) leaves every weight free and produces noisier, denser filters; $\gamma = 10^{-2}$ (strong_reg) erodes both filter sharpness and the energy barrier, because most weights have been driven to zero and the model can no longer represent the digit basins. The baseline $\gamma = 10^{-3}$ sits at the sweet spot between these two regimes.

(iv) *Capacity and CD depth.* Doubling L from 12 to 24 increases the energy barrier by $\sim 25\%$ and lowers the final gradient RMS, the clearest single-knob improvement in the sweep. Halving capacity ($L=6$) forces each unit to absorb more variance, raising filter sharpness but shrinking the basin separation. CD depth has a smaller effect than naive intuition would suggest: $N_t = 1$ already produces sensible filters; the benefit of $N_t = 5$ shows up mainly as a slightly lower final RMS, because a longer chain is a better – but still biased – estimate of the model expectation.

An additional dedicated L study (RMSprop, $\gamma = 10^{-3}$, $N_t = 2$, 50 epochs) confirms the capacity trend: the converged $\text{std}(g_W)$ scales as $1.4 \times 10^{-2} \rightarrow 1.5 \times 10^{-2} \rightarrow 1.1 \times 10^{-2}$ for $L = 6, 12, 24$, with the largest model also giving the highest free-energy barrier. The price paid for $L = 24$ is that some units become dead or duplicate (Fig. 1 already shows one such unit at $L = 12$): in a strict generative-

modelling regime (maximising ΔF between basins) the data favour $L = 24$, while $L = 12$ is the best balance of capacity, interpretability, and compute budget.

4. CONCLUSIONS

We have used a single RBM trained on binarised MNIST $\{0, 1, 2\}$ to verify, on the same model, the four mechanisms that govern its learning dynamics. The gradient RMS approaches a $1/\sqrt{N}$ noise floor that the mini-batch schedule is explicitly designed to lower; Hinton’s visible-bias initialisation matches the model marginal to the data marginal before any weight has been updated; the trained model carves free-energy basins separated by barriers $\Delta F \sim 60\text{--}100$ ($\text{SNR} \sim 1.1\text{--}1.8$) that suppress spontaneous CD class-mixing by tens of orders of magnitude; and the hyperparameter sweep ranks the hidden width L , the ℓ_1 weight γ and the number of CD steps N_t as the dominant knobs, with RMSprop strongly favoured over SGD because it absorbs the heteroscedastic CD noise without tuning.

Three takeaways go beyond the class material. (1) The natural definition of RBM convergence is not the gradient itself but its ratio to the $1/\sqrt{N}$ floor; training past unity is wasted unless N is increased. (2) Configurations producing visually indistinguishable digits (e.g. baseline vs. no_reg) can differ by $\sim 30\%$ in ΔF , so we propose the dimensionless ratio $\text{SNR} = \Delta F / \sigma_{\text{within}}$ as a single-number quality metric: closed-form from Eq. (4), invariant under uniform energy rescaling, and predictive of Gibbs mixing time. (3) The phase-transition picture suggests an obvious mitigation: replace the pixel-flip path by a string-method geodesic and combine with parallel tempering, to escape the basins behind the quasi-ergodicity of Fig. 4.

Project authorship. The code, experiments, analysis, figures, and report were completed by Amirmohammad Saiedi Saber.

REFERENCES

- [1] P. Mehta, M. Bukov, C.-H. Wang, A. G. R. Day, C. Richardson, C. K. Fisher and D. J. Schwab, *Physics Reports* **810**, 1–124 (2019).
- [2] M. Bortoletto, *Study of performances for Restricted Boltzmann Machines*, Final dissertation, Università degli Studi di Padova, Dipartimento di Fisica e Astronomia “Galileo Galilei” (2021).
- [3] G. E. Hinton, in *Neural Networks: Tricks of the Trade, Second Edition*, edited by G. Montavon, G. B. Orr and K.-R. Müller (Springer, 2012) pp. 599–619.